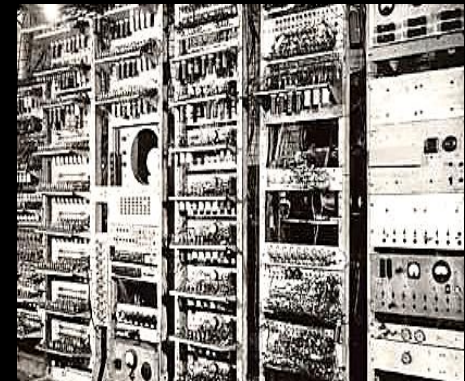


THE EVOLUTION Of COMPUTER



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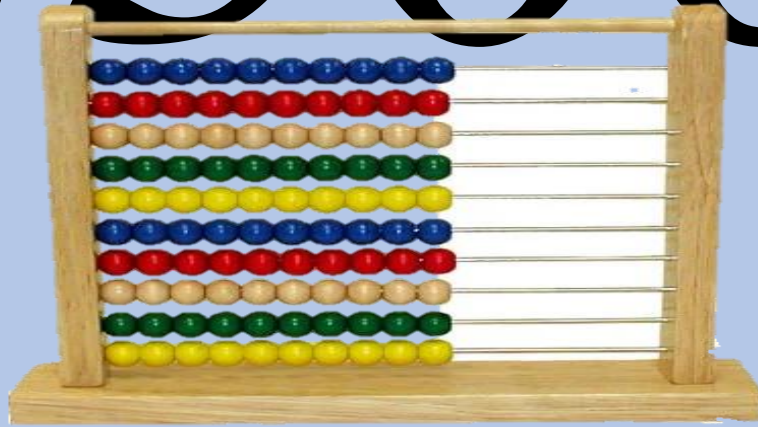


Introduction:

Computers are remarkable machines that have revolutionized the way we live, work, and communicate. At their core, computers are devices capable of processing data and performing a wide range of tasks with incredible speed and accuracy. They operate by following instructions encoded in software, which allows them to execute complex calculations, store vast amounts of information, and automate processes that were once manual. From the early mechanical calculators to the modern-day supercomputers, the evolution of computers has been driven by advancements in technology, making them more powerful, compact, and versatile.

In this presentation, you will be able to learn about the evolution of computers.

2300 BC



The Abacus

The abacus is an ancient calculating tool, often considered the first computer. It consists of a frame with rods and beads, allowing users to perform arithmetic operations like addition and subtraction manually. Originating around 2300 BCE, the abacus laid the groundwork for modern computation by providing a systematic way to handle numerical calculations.

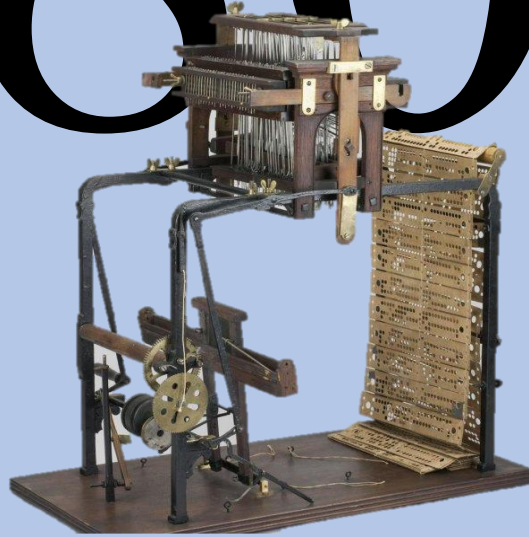
1642



The Pascaline

The first calculating machine, known as the Pascaline, was built in 1642 by the French physicist **Blaise Pascal** when he was 19 years old. This early ancestor of the pocket calculator counts whole numbers by the cogs, or teeth, on a cogwheel. The cogwheel is connected to a mechanical gear system.

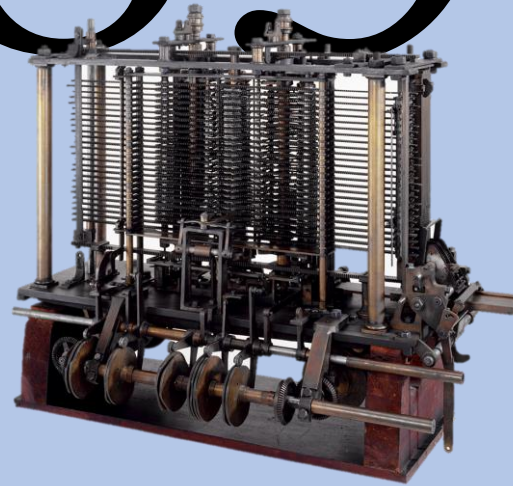
1801



The Jacquard Loom

The Jacquard mechanism, invented by Frenchman **Joseph Marie Jacquard** and first demonstrated in 1801, simplified the way in which complex textiles such as damask were woven. The mechanism involved the use of thousands of punch cards laced together. Each row of punched holes corresponded to a row of a textile pattern.

1837



The Analytical Engine

The Analytical Engine, designed by **Charles Babbage** in the 19th century, is generally considered the first computer. Initially inspired by his earlier work on the simpler Difference Engine, Babbage envisioned a more advanced, general-purpose computing machine. The Analytical Engine was designed to be fully program-controlled and capable of performing any calculation. Its design included four key components: the mill (analogous to the modern CPU), the store (equivalent to memory and storage), and the reader and printer (serving as input and output devices). These components form the foundation of all modern computers..

1941



The Z3

The Z3, an early computer built by German engineer **Konrad Zuse** working in complete isolation from developments elsewhere, uses 2,300 relays, performs floating point binary arithmetic, and has a 22-bit word length. The Z3 was used for aerodynamic calculations but was destroyed in a bombing raid on Berlin in late 1943. Zuse later supervised a reconstruction of the Z3 in the 1960s, which is currently on display at the Deutsches Museum in Munich

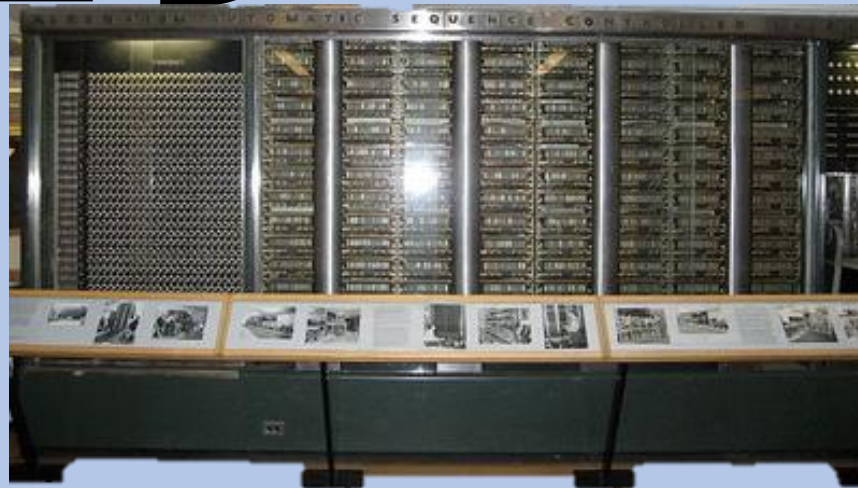
1943



The Colossus

The Colossus was developed in 1943 by **engineer Tommy Flowers**, based on plans by the **mathematician Max Newman**. It was designed to decode the encrypted transmissions from the German teleprinter **Lorenz cipher**.

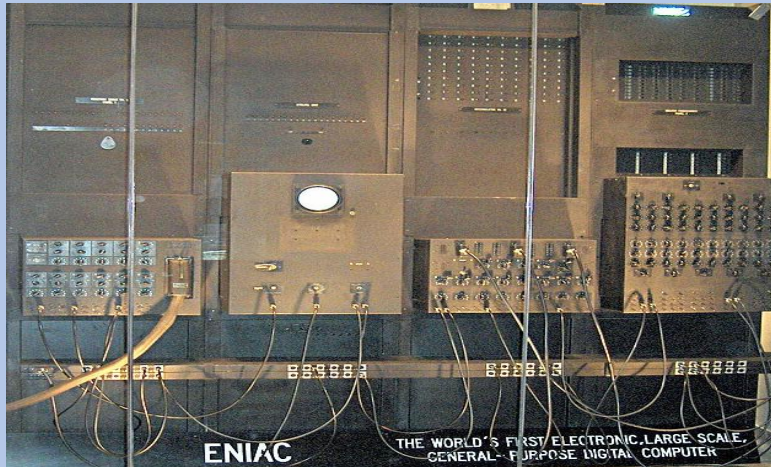
1944



The Harvard Mark I

The Harvard Mark I computer, also known as the IBM Automatic Sequence Controlled Calculator (ASCC), was the first electromechanical computer capable of automated complex calculations. It weighed five tons, measured over 50 feet long, and was comprised of approximately 750,000 parts.. It was used in the war effort during the last part of World War II. One of the first programs to run on the Mark I was initiated on 29 March 1944 by **John von Neumann**.

1945



The ENIAC

Electronic Numerical Integrator and Computer, the first programmable general-purpose electronic digital computer, was built during World War II in the United States by **John Mauchly, J. Presper Eckert, Jr.**, and **their team at the University of Pennsylvania**. Funded by the government and developed under the army's direction, work on ENIAC began in 1943, with mathematician **John von Neumann** joining the project in 1944 as a consultant. Although designed specifically for calculating artillery range tables, ENIAC had limitations as a universal computer. It used plugboards to program instructions, enabling fast electronic processing, but required days to rewire for new tasks, making it only somewhat programmable.

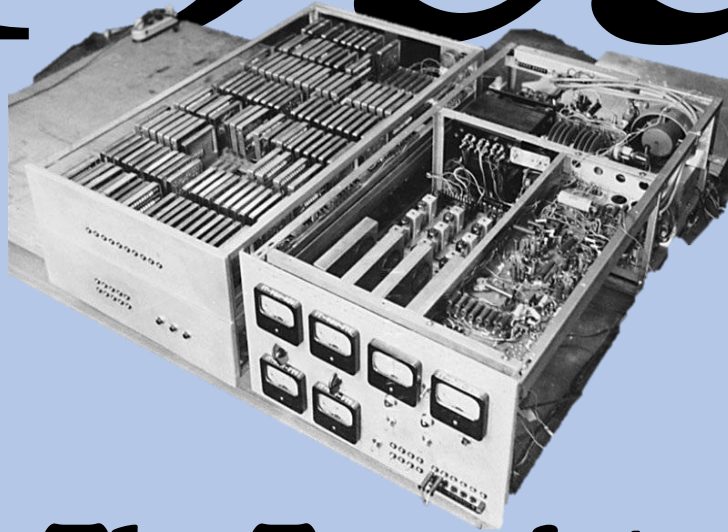
1951



The UNIVAC

UNIVAC stands for **Universal Automatic Computer** which was the world's first computer, and the first commercial computer employing transistorised circuits. It was developed by a group of engineers from the Eckert-Mauchly Computer Corporation (EMCC), led by **Ted C. J**

1955

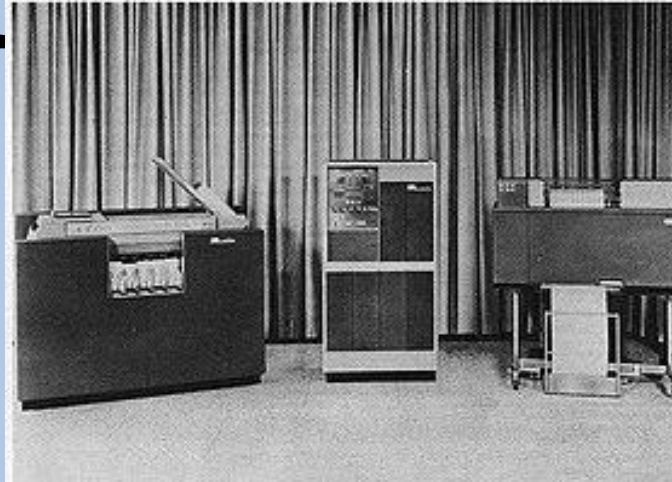


The Transistors

Transistors replaced vacuum tubes and ushered in the second generation of computers. The transistor was invented in 1947 but did not see widespread use in computers until the late 1950s. The transistor was far superior to the vacuum tube, allowing computers to become smaller, faster, cheaper, more energy-efficient and more reliable than their first-generation predecessors. Though the transistor still generated a great deal of heat that subjected the computer to damage, it was a vast improvement over the vacuum tube. Second-generation computers still relied on punched cards for input and printouts for output.

Second Generation Computers

1959



The IBM 1401

The IBM 1401 is a variable-wordlength decimal computer that was announced by IBM on October 5, 1959. The first member of the highly successful IBM 1400 series, it was aimed at replacing unit record equipment for processing data stored on punched cards and at providing peripheral services for larger computers.

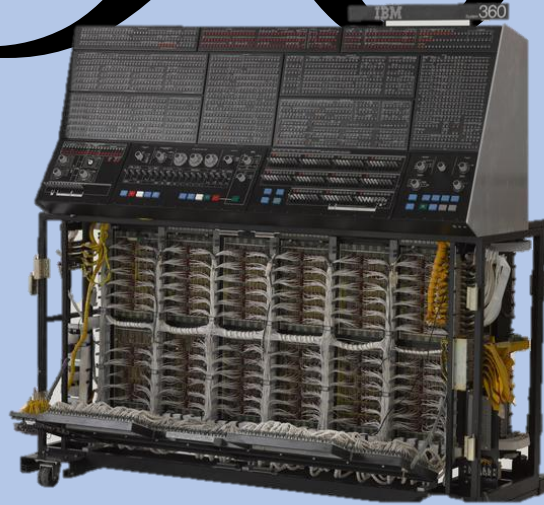
1958



The Integrated Circuit

An IC is the fundamental building block of all modern electronic devices. As the name suggests, it's an integrated system of multiple miniaturized and interconnected components embedded into a thin substrate of semiconductor material (usually silicon crystal)

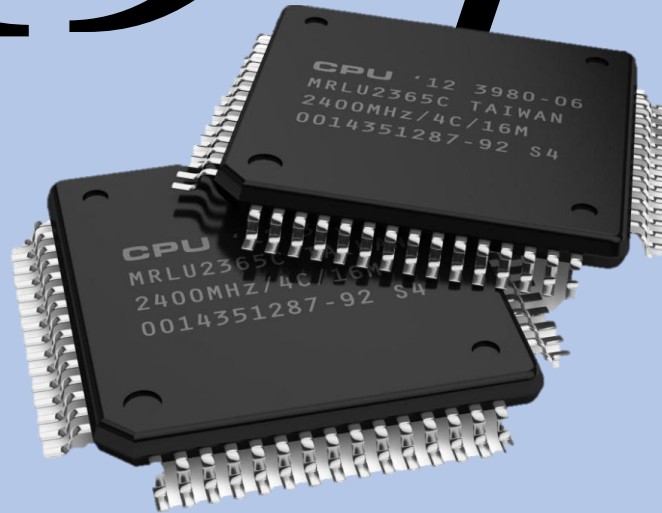
1964



The IBM System /360

The IBM System/360 is a family of mainframe computer systems that was announced by IBM on April 7, 1964, and delivered between 1965 and 1978. It was the first family of computers designed to cover both commercial and scientific applications and a complete range of applications from small to large.

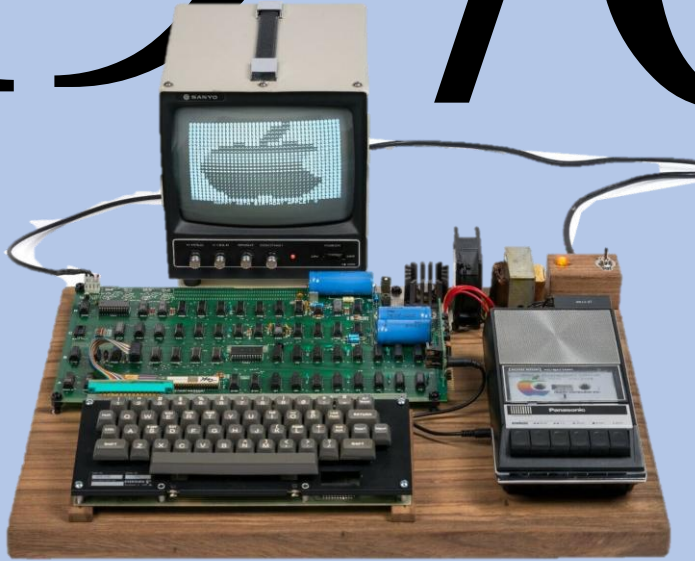
1971



The Microprocessors

microprocessor, any of a type of miniature electronic device that contains the arithmetic, logic, and control circuitry necessary to perform the functions of a digital computer's central processing unit

1976



The Apple I

The Apple Computer 1, later known predominantly as the Apple I, is an 8-bit motherboard-only personal computer designed by Steve Wozniak and released by the Apple Computer Company in 1976.

1977



The Apple II

The Apple II series of microcomputers was initially designed by Steve Wozniak, manufactured by Apple Computer, and launched in 1977 with the Apple II model that gave the series its name. It was followed by the Apple II Plus, Apple IIe, Apple IIc, and Apple IIc Plus, with the 1983 IIe being the most popular

1981



The IBM PC

IBM's brand recognition, along with a massive marketing campaign, ignites the fast growth of the personal computer market with the announcement of its own personal computer (PC). The first IBM PC, formally known as the IBM Model 5150, was based on a 4.77 MHz Intel 8088 microprocessor and used Microsoft's MS-DOS operating system. The IBM PC revolutionized business computing by becoming the first PC to gain widespread adoption by industry. The IBM PC was widely copied ("cloned") and led to the creation of a vast "ecosystem" of software, peripherals, and other commodities for use with the platform.

1984



The Macintosh

Apple introduces the Macintosh with a television commercial during the 1984 Super Bowl, which plays on the theme of totalitarianism in George Orwell's book 1984. The ad featured the destruction of "Big Brother" – a veiled reference to IBM -- through the power of personal computing found in a Macintosh. The Macintosh was the first successful mouse-driven computer with a graphical user interface and was based on the Motorola 68000 microprocessor. Its price was \$2,500. Applications that came as part of the package included MacPaint, which made use of the mouse, and MacWrite, which demonstrated WYSIWYG (What You See Is What You Get) word processing.

1990-BEYOND

The Fifth Generation Computers

The computers of fifth-generation use AI (artificial intelligence) technology that includes: Development of expert systems, Game Playing, Robotics, Natural language understanding, and Neural Networks. The AI technology made these computers to understand human language as well as recognize graphs and pictures.

CONCLUSION:

Computers have evolved from simple mechanical calculators to advanced digital machines, becoming essential tools in modern society. From the pioneering designs of the Analytical Engine to the revolutionary ENIAC, computers have transformed how we process information, solve problems, and interact with the world. Their continuous advancement shapes our future, driving innovation across all sectors.